2022 Minnesota Academic Standards in Mathematics
Kindergarten

This official standards document contains the mathematics standards revised in 2022 and put into rule.

Introduction

The Minnesota K–12 Academic Standards in Mathematics represent the work of the Mathematics Standards Review Committee. This committee included K–12 teachers, administrators, college faculty and representatives of educational and community organizations. The committee membership, timeline and assumptions that guided their work are found on the Minnesota Department of Education (MDE) Mathematics webpage, (https://education.mn.gov/MDE/dse/stds/Math/) or scan the QR code to the right.

During the 2021–22 school year, the Mathematics Standards Review Committee reviewed the 2007 Minnesota K–12 Academic Standards in Mathematics, other states' recently revised standards, current academic research, K–12 instructional best practices and public feedback. This review process followed the guidelines in Minnesota Statutes 2023, section 120B.021, subdivision 4e. The standards and anchor standards provide a summary description of student learning. The kindergarten through grade 12 benchmarks identify a "specific knowledge or skill that a student must master to complete part of an academic standard by the end of the grade level or grade band" (Minnesota Statutes 2023, section 120.B.018).

Minnesota Statutes 2022, section 120B.021, requires that there be statements of standards and benchmarks. Minnesota Statutes 2022, section 120B.021, subdivision 4a states that "the commissioner must include the contributions of Minnesota American Indian tribes and communities as related to the academic standards during the review and revision of the required academic standards." The four Dakota and seven Anishinaabe/Ojibwe Tribal Nations, as well as Minnesota's significant Urban Indigenous communities, have been relatively invisible in Minnesota's academic standards, and Minnesota Statutes 120B.021 counteracts that invisibility and lack of representation in Minnesota's education system. This statute requirement, added by the legislature in 2007, demonstrates the commitment of the State of Minnesota to ensure that the contributions of the Tribal Nations in Minnesota are integrated into student academic learning and instruction. Minnesota's K–12 Academic Standards in Mathematics include the historical and current contributions of the eleven sovereign, federally recognized Tribal Nations in Minnesota and Minnesota's Urban Indigenous communities.

Career, College and Community Readiness Vision Statement

"Let's put our minds together and see what kind of future we can build for our children" (Tatanka Iyotake [Chief Sitting Bull], 1868).

Mathematics belongs to all of us. Mathematics is part of our everyday lives and is rooted in all cultures. Equitable access to mathematics for each and every student is accomplished by recognizing, respecting and attending to the diversity of our students with the aim of ensuring that all students are able to learn and thrive.

The goal of mathematics education is to prepare each and every student for effective participation in society, including in their career(s), post-secondary education and daily decision-making about everything from finances, personal health, civic discourse and policy-making to their ability to comprehend and analyze data. All students should learn mathematics "in order to expand professional opportunities, understand and critique the world, and experience the joy, wonder and beauty of mathematics" (National Council of Teachers of Mathematics [NCTM], 2018).

Mathematics education needs to evolve with the constant cultural and technological changes in our society. The importance of math education, related to career, college and community readiness, is supported by the World Economic Forum's *The Future of Jobs Report* (World Economic Forum, 2022). The report lists complex problem-solving, critical thinking, reasoning, analytical thinking and active learning in its list of Top Ten Workplace Skills. These are all skills that are developed in the study of mathematics throughout a student's K–12 experience.

Mathematically literate students have the experiences, mindset, knowledge and skills to be career and college ready and engage as productive members in their community. They are empowered to use mathematics as a resource to successfully navigate pathways towards achieving their aspirations (MDE Career and College Readiness [CCR] Resource Guide).

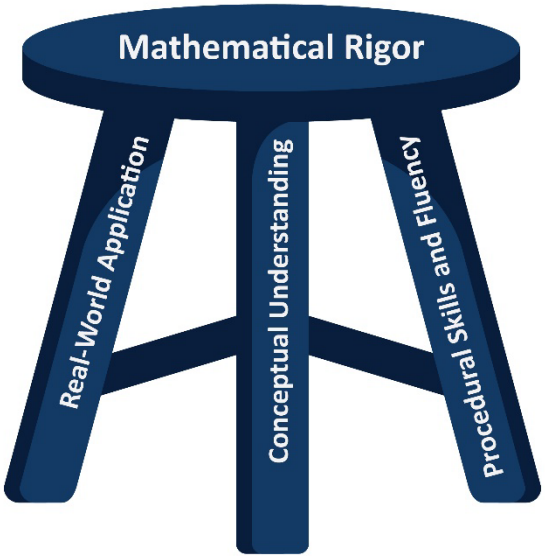
Minnesota career, college and community ready students will:

- Be curious, pose questions and seek patterns in order to make sense of their world.
- Communicate their mathematical thinking and contribute to high-level math discussions.
- Be persistent, flexible, collaborative and creative problem solvers.
- Make connections between mathematics concepts and other disciplines, experiences outside the classroom, interests and career aspirations, as well as the connections amongst mathematical ideas.

- Build conceptual understanding, thinking and reasoning in order to develop procedural fluency and flexible problem-solving strategies.
- Collaborate with cultural perspectives and traditions like and unlike one's own, allowing students to make sense of mathematical concepts and value various mathematical identities connected to lived experiences.
- Solve problems connected to place, story, cultural practices, language and perspectives relevant to historical and contemporary Dakota and Anishinaabe/Ojibwe communities (Minnesota Statutes 2021, section 120B.021).

In order to support students to become career, college and community ready, these standards and benchmarks:

- Pursue mathematical rigor with an equal intensity of conceptual understanding, application and procedural skill and fluency (Funderburk et al., 2016).
- Incorporate the eight Standards for Mathematical Practice (SMPs) to promote experiences that empower students to be "confident in themselves as doers, knowers, and sense makers of mathematics" (NCTM, 2020).
- Incorporate five contexts in Dimension 2 to use as a lens for the teaching and learning of the standards and benchmarks.
- Equip students to analyze information in a data-rich environment by including data science and computer science, including computational thinking, throughout the K–12 mathematics experience.
- Empower students to make informed financial decisions now and into the future. Students will learn mathematics which, when combined with the Minnesota K–12 Social Studies Standards, addresses the five key components of financial literacy: earn, spend, save and invest, borrow and protect.



Organization of the Standards

The organization and structure of the 2022 Minnesota K–12 Academic Standards in Mathematics communicate how mathematics education is conceptualized for Minnesota students. The standards are ordered by grade, beginning in kindergarten (represented by zero) and ending in high school (represented by 9). In each grade the standards are subdivided into strand, anchor standard and benchmark. Additional dimensions within benchmarks are mathematical practices and benchmark contexts, which serve to encourage integration of concepts through process and context.

The three strands—Data and Probability, Spatial Reasoning and Patterns and Relationships—organize the standards. Most grades have seven anchor standards that are consistent across kindergarten through grade 11. Each standard will have one or more benchmarks that are grade-level specific that define a specific knowledge or skill a student must master. The benchmarks are placed at the end of the grade level where mastery is expected, with the recognition that a progression of learning experiences in earlier grades builds the foundation for mastery later on.

Strands

The standards are organized into three strands common across all grade levels:

1. **Data and Probability** – *"Data are not merely numbers but numbers in context"* (Cobb and Moore, 1997). Students will develop questions about situations impacting their everyday lives, make a plan to produce data and provide answers by organizing, describing and summarizing the data. In addition, students will develop and evaluate inferences and make predictions. They will find patterns and order from random phenomena to analyze and understand important relationships and use the concept of variability to make sense of the world.
2. **Spatial Reasoning** – Students will actively make sense of relationships between and within geometric figures, generalize statements about the figures and develop arguments about these relationships.
3. **Patterns and Relationships** – Mathematics is the science of pattern and order. Students will organize and describe the world using representations of numbers and operations. They will learn actions that transform numbers as well as ways of thinking that bring them back to where they started. Students will analyze and describe relationships among variables by focusing on how variables change and stay the same.

Anchor Standards

Minnesota Statutes 2021, section 120B.021, requires that there be statements of standards and benchmarks. Anchor standards are a summary description of student learning that reflects a learning progression, spanning from kindergarten to graduation. There are seven anchor strands organized into three strands.

Anchor Standards by Strand

Strand 1: Data and Probability

1. Data Sciences: Identify, formulate and investigate statistical questions by collecting data considering cultural perspectives, analyzing and interpreting data and communicating the results.

2. Chance and Uncertainty: Apply and explain the concepts of probability to interpret data, generate questions, predict and make informed decisions to solve problems and communicate ideas.

Strand 2: Spatial Reasoning

3. Measurement: Investigate measurement using a variety of tools, units, systems, processes and techniques in various cultures. Explain and reason with attributes, estimations and formulas to communicate measurement(s) and relationships effectively. Justify decisions and consider the reasonableness of the measurement.
4. Geometry: Analyze characteristics of geometric shapes to make mathematical arguments and justifications about geometric relationships. Use visualization and geometric modeling to compare, solve problems and communicate ideas.

Strand 3: Patterns and Relationships

5. Number Relationships: Describe, Interpret and use quantities, relationships between quantities, representations of quantities, and number systems. Describe operations and the relationship between operations. Use strategies and procedures accurately, efficiently and flexibly. Assess the reasonableness of the results.
6. Equivalence and Relational Thinking: Use concepts and properties of equivalence and relational thinking to represent and compare numerical expressions, proportional relationships, algebraic expressions and equations.
7. Patterns and Relationships: Represent and connect mathematical patterns and relationships using verbal descriptions, generalizations, tables and graphs. Use representations to generate questions, make predictions and solve mathematical problems.

Benchmarks

A benchmark is a "specific knowledge or skill that a student must master to complete part of an academic standard by the end of the grade level or grade band" (Minnesota Statutes 2013, section 120B.018, subdivision 3). The benchmarks are placed at the grade level where mastery is expected, and with recognition that the progression of learning experiences in earlier grades builds the foundation for mastery later on. The benchmarks include notations of two cross-cutting dimensions described below. The benchmarks inform the graduation requirements for students, which read: *"three credits of mathematics, including an algebra II credit or its equivalent, sufficient to satisfy all of the academic standards in mathematics;" and "an algebra I credit by the end of 8th grade sufficient to satisfy all of the 8th grade*

standards in mathematics" (Minnesota Statutes 2019, section120B.024, subdivision 1).[1] The standards and benchmarks for grades 8 and 9–11 define the completion and mastery of Algebra 1 in grade 8 and Algebra 2 in grades 9–11.

Mathematical Dimensions Description and Notation

Practices used by mathematicians and authentic contexts for mathematical thinking serve as dimensions that interact with the grade level benchmarks. Standards of Mathematical Process and benchmark contexts are corresponding cross-cutting dimensions that are referenced in each benchmark. The dimensions encourage integration of benchmarks with process (dimension 1) and contributions and context (dimension 2). Mathematical connections to a coherent/comprehensive view of the world provide students access to Minnesota's Career, College and Community Readiness goal.

Dimension 1: Standards of Mathematical Practice (MP)

This dimension focuses on the Standards for Mathematical Practice.[2] These practices describe the processes, behaviors and habits of mind that are exhibited by students who are mathematically proficient. Mathematical understanding is the intersection of these practices and mathematics content. It is critical that the Standards for Mathematical Practice are embedded in daily mathematics instruction. The mathematical practices are:

MP1: Make sense of problems and persevere in solving them.
MP2: Reason abstractly and quantitatively.
MP3: Construct viable arguments and critique the reasoning of others.
MP4: Model with mathematics.
MP5: Use appropriate tools strategically.
MP6: Attend to precision.
MP7: Look for and make use of structure.
MP8: Look for and express regularity in repeated reasoning.

[1] This 2019 citation represents the statute and its wording at the time of the committee work. The current statute, updated in 2023, shifts this specificity to 120B.021, subdivision 1.
[2] Appendix 1

Dimension 2: Benchmark Contexts

This dimension focuses on using context to do and learn mathematics. All students should make sense of and persevere in mathematical problem-solving experiences using local contexts of the learning community, within the context of Minnesota and in the contexts of our nation and the global community. The benchmarks noted with the symbols below represent opportunities to utilize the named contexts to teach the mathematics within the benchmark and standard.

- ✤ **Minnesota Tribal Nations Contexts:** Connect mathematical problem-solving experiences and contributions to place, story, cultural practices, language and perspectives relevant to historical and contemporary Dakota and Anishinaabe/Ojibwe communities. The four directions symbol (✤) indicates benchmarks' instructional topics that could include contributions of Dakota and Anishinaabe/Ojibwe in current-day Minnesota.

- $ **Financial Literacy Contexts:** Allow students to make sense of mathematics through the context of money to manage one's financial resources effectively for a lifetime of financial security. The dollar sign ($) represents financial literacy contexts.

- # **Computer Science Contexts:** Computer science is an independent field of study that focuses on learning how to think computationally and adapting/creating new technology. There are many connections with mathematics, and integrating the subjects can deepen student understanding in mathematics and provide more students with access to computer science. The number sign (#) represents computer science contexts.

- μ **Modeling Contexts:** Students apply mathematical problem-solving experiences to connect with the Standards of Mathematical Practices meriting greater time, resources, innovative energies and focus to deeply explore the content of the benchmark. The mu symbol (μ) represents modeling contexts.

- ☼ **Real-world Contexts:** Apply mathematical problem-solving experiences to real-world situations in contexts connected to students' lived experiences, cultural perspectives and traditions. The sun symbol (☼) represents real-world contexts.

How to Read the Standards, Benchmarks and Dimensions

Each benchmark has a 4-digit code, which includes the grade, strand, standard and benchmark numbers. See the examples below.

Example 1: Benchmark 0.2.4.3

- The 0 represents the kindergarten grade level.
- The 2 represents the second strand: Spatial Reasoning.
- The 4 represents the fourth anchor standard: Analyze characteristics of geometric shapes to make mathematical arguments and justifications about geometric relationships. Use visualization and geometric modeling to compare, solve problems and communicate ideas.
- The 3 represents the third benchmark: Compose, decompose and name simple shapes. Recognize shapes regardless of their overall size and orientation. (MP1, MP2) µ

Grade	Strand	Anchor Standard	Code	Benchmark
K	Spatial Reasoning	Geometry: Analyze characteristics of geometric shapes to make mathematical arguments and justifications about geometric relationships. Use visualization and geometric modeling to compare, solve problems and communicate ideas.	0.2.4.3	Compose, decompose and name simple shapes. Recognize shapes regardless of their overall size and orientation. (MP1, MP2) µ

Example 2: Benchmark 9.3.7.10

- The 9 represents the high school, 9–11 grade band.
- The 3 represents the third strand: Patterns and Relationships
- The 7 represent**s** the seventh Anchor Standard: Represent and connect mathematical patterns and relationships using verbal descriptions, generalizations, tables and graphs. Use representations to generate questions, make predictions and solve mathematical problems.
- The 10 represents the 10th benchmark: Use the concept of a function as a connection between inputs and outputs to find function values and use function notation.

Grade	Strand	Anchor Standard	Code	Benchmark
9	Patterns and Relationships	Patterns and Relationships: Represent and connect mathematical patterns and relationships using verbal descriptions, generalizations, tables and graphs. Use representations to generate questions, make predictions and solve mathematical problems.	9.3.7.10	Use the concept of a function as a connection between inputs and outputs to find function values and use function notation. (MP2)

Mathematics Standards Grade-Level Tables

Kindergarten

Strand 1: Data and Probability

Grade	Strand	Anchor Standard	Code	Benchmark
K	Data and Probability	Data Sciences: Identify, formulate and investigate statistical questions by collecting data, considering cultural perspectives, analyzing and interpreting data and communicating the results.	0.1.1.1	Notice and describe patterns in data-rich situations. (MP1, MP7) ✣ # ☼
K	Data and Probability	Data Sciences: Identify, formulate and investigate statistical questions by collecting data, considering cultural perspectives, analyzing and interpreting data and communicating the results.	0.1.1.2	Organize objects, draw pictures or use tally marks to represent data and communicate observations. (MP3, MP6) # µ

Strand 2: Spatial Reasoning

Grade	Strand	Anchor Standard	Code	Benchmark
K	Spatial Reasoning	Measurement: Investigate measurement using a variety of tools, units, systems, processes and techniques in various cultures. Explain and reason with attributes, estimations and formulas to communicate measurement(s) and relationships effectively. Justify decisions and consider the reasonableness of the measurement.	0.2.3.1	Compare objects with a measurable attribute in common, to see which object has “more of,” “less of” or the “same as” the attribute and explain the reasoning. (MP3, MP5) ✣ $ ☼

Grade	Strand	Anchor Standard	Code	Benchmark
K	Spatial Reasoning	Measurement: Investigate measurement using a variety of tools, units, systems, processes and techniques in various cultures. Explain and reason with attributes, estimations and formulas to communicate measurement(s) and relationships effectively. Justify decisions and consider the reasonableness of the measurement.	0.2.3.2	Describe several measurable attributes of objects such as length and weight. (MP4, MP6) ✣ ☼
K	Spatial Reasoning	Geometry: Analyze characteristics of geometric shapes to make mathematical arguments and justifications about geometric relationships. Use visualization and geometric modeling to compare, solve problems and communicate ideas.	0.2.4.1	Sort objects using characteristics such as shape, size, color and thickness. (MP1, MP3) ✣ ☼
K	Spatial Reasoning	Geometry: Analyze characteristics of geometric shapes to make mathematical arguments and justifications about geometric relationships. Use visualization and geometric modeling to compare, solve problems and communicate ideas.	0.2.4.2	Identify and compare two- and three-dimensional shapes such as squares, circles, triangles, rectangles, trapezoids, hexagons, cubes, cones, cylinders and spheres using informal language to describe their similarities, differences, parts and other attributes. (MP2) ✣ ☼
K	Spatial Reasoning	Geometry: Analyze characteristics of geometric shapes to make mathematical arguments and justifications about geometric relationships. Use visualization and geometric modeling to compare, solve problems and communicate ideas.	0.2.4.3	Compose, decompose and name simple shapes. Recognize shapes regardless of their overall size and orientation. (MP1, MP2) µ

Grade	Strand	Anchor Standard	Code	Benchmark
K	Spatial Reasoning	Geometry: Analyze characteristics of geometric shapes to make mathematical arguments and justifications about geometric relationships. Use visualization and geometric modeling to compare, solve problems and communicate ideas.	0.2.4.4	Describe objects in the environment using names of shapes. Describe the relative positions of these objects using terms such as above, below, beside, in front of, behind and next to. (MP1, MP6) ✣ ☼

Strand 3: Patterns and Relationships

Grade	Strand	Anchor Standard	Code	Benchmark
K	Patterns and Relationships	Number Relationships: Describe, Interpret and use quantities, relationships between quantities, representations of quantities, and number systems. Describe operations and the relationship between operations. Use strategies and procedures accurately, efficiently and flexibly. Assess the reasonableness of the results.	0.3.5.1	Recognize that a number can be used to represent how many objects are in a set or to represent the position of an object in a sequence. When counting objects, say the number names in the standard order, pairing each object with one and only one number name and each number with one and only one object. Understand that the last number said tells the number of objects counted. Understand that each successive number refers to a quantity that is one more. Name the position of an object in a sequence (ordinal count). (MP1, MP6) ✣ ☼

Grade	Strand	Anchor Standard	Code	Benchmark
K	Patterns and Relationships	Number Relationships: Describe, Interpret and use quantities, relationships between quantities, representations of quantities, and number systems. Describe operations and the relationship between operations. Use strategies and procedures accurately, efficiently and flexibly. Assess the reasonableness of the results.	0.3.5.2	Count collections of objects up to 31 by grouping in 10s using ten-frames, cups or other tools. (MP6, MP7) ✜ $ ☼
K	Patterns and Relationships	Number Relationships: Describe, Interpret and use quantities, relationships between quantities, representations of quantities, and number systems. Describe operations and the relationship between operations. Use strategies and procedures accurately, efficiently and flexibly. Assess the reasonableness of the results.	0.3.5.3	Read, write, compare, order and represent whole numbers from 0 to at least 31 (with 0 representing the count of no objects) to answer the question “how many?”. Representations may include numerals, pictures, real objects, picture graphs, spoken words and manipulatives, such as connecting cubes. The numbers from 11 to 19 are composed of a 10 and one, two, three, four, five, six, seven, eight or nine ones. (MP4, MP8) ✜
K	Patterns and Relationships	Number Relationships: Describe, Interpret and use quantities, relationships between quantities, representations of quantities, and number systems. Describe operations and the relationship between operations. Use strategies and procedures accurately, efficiently and flexibly. Assess the reasonableness of the results.	0.3.5.4	Count forward, with and without objects, to at least 31. Count backward from 20. (MP6) ☼

Grade	Strand	Anchor Standard	Code	Benchmark
K	Patterns and Relationships	Number Relationships: Describe, Interpret and use quantities, relationships between quantities, representations of quantities, and number systems. Describe operations and the relationship between operations. Use strategies and procedures accurately, efficiently and flexibly. Assess the reasonableness of the results.	0.3.5.5	Find a number that is 1 more or 1 less than a given number. (MP7, MP8)
K	Patterns and Relationships	Number Relationships: Describe, Interpret and use quantities, relationships between quantities, representations of quantities, and number systems. Describe operations and the relationship between operations. Use strategies and procedures accurately, efficiently and flexibly. Assess the reasonableness of the results.	0.3.5.6	Solve and represent a variety of addition and subtraction contextual situation types using objects, drawings, mental images or equations within 10. (MP4, MP5) $ µ
K	Patterns and Relationships	Number Relationships: Describe, Interpret and use quantities, relationships between quantities, representations of quantities, and number systems. Describe operations and the relationship between operations. Use strategies and procedures accurately, efficiently and flexibly. Assess the reasonableness of the results.	0.3.5.7	Compose and decompose numbers less than or equal to 10 into pairs in more than one way with objects and pictures. Record each decomposition with a drawing or equation. (MP7)
K	Patterns and Relationships	Number Relationships: Describe, Interpret and use quantities, relationships between quantities, representations of quantities, and number systems. Describe operations and the relationship between operations. Use strategies and procedures accurately, efficiently and flexibly. Assess the reasonableness of the results.	0.3.5.8	Fluently add and subtract within 5. (MP2)

Grade	Strand	Anchor Standard	Code	Benchmark
K	Patterns and Relationships	Equivalence and Relational Thinking: Use concepts and properties of equivalence and relational thinking to represent and compare numerical expressions, proportional relationships, algebraic expressions and equations.	0.3.6.1	Identify whether the number of objects in one group is greater than, less than or equal to the number of objects in another group (by using matching, counting strategies and a number line). (MP2, MP5) ☼
K	Patterns and Relationships	Equivalence and Relational Thinking: Use concepts and properties of equivalence and relational thinking to represent and compare numerical expressions, proportional relationships, algebraic expressions and equations.	0.3.6.2	Recognize that the equal sign (=) is a comparison symbol of two math expressions of equal value number. (MP6)
K	Patterns and Relationships	Patterns and Relationships: Represent and connect mathematical patterns and relationships using verbal descriptions, generalizations, tables and graphs. Use representations to generate questions, make predictions and solve mathematical problems.	0.3.7.1	Recognize, create, complete and extend simple patterns using shape, color, size, number, sounds and movements. Patterns may be repeating, growing or shrinking. (MP1, MP7) ✜ # ☼
K	Patterns and Relationships	Patterns and Relationships: Represent and connect mathematical patterns and relationships using verbal descriptions, generalizations, tables and graphs. Use representations to generate questions, make predictions and solve mathematical problems.	0.3.7.2	Recognize patterns in counting. Skip count by 10s starting at zero up to 100. (MP7) ✜ $ ☼

References

Cobb, G. W., and Moore, D. S. (1997). Mathematics, statistics, and teaching. *The American Mathematical Monthly*, *104*(9), 801. https://www.jstor.org/stable/2975286

Funderburk, J., Fann, J., Alberti, S., and Greenstein, M. (2016, March). *Why Rigor Doesn't Mean Harder*. *Achieve the Core*.

The National Council of Teachers of Mathematics, Inc. (n.d.). Page 23. In *Catalyzing Change in Early Childhood and Elementary Mathematics*. Essay, NCTM.

World economic forum. (n.d.). Retrieved October 5, 2022. https://www3.weforum.org/docs/WEF_Future_of_Jobs_2020.pdf.

The National Council of Teachers of Mathematics, Inc. (2018). Catalyzing Change: Next Steps. In *Catalyzing change in high school mathematics initiating critical conversations*. Essay. Retrieved from https://www-nctm-org.webpkgcache.com/doc/-/s/www.nctm.org/uploadedFiles/Standards_and_Positions/executive%20summary.pdf

Minnesota Department of Education, MDE Career and College Readiness (CCR) Resource Guide. Retrieved from https://education.mn.gov/mde/dse/ccs/.

"50th Anniversary Celebration." Sitting Bull College. Accessed November 30, 2023. https://www.sittingbull.edu/news/50th-anniversary-celebration/.

Appendix 1

Standards for Mathematical Practice

Math Practice 1: Make sense of problems and persevere in solving them.

Mathematically proficient students start by explaining to themselves the meaning of a problem and looking for entry points to its solution. They analyze givens, constraints, relationships and goals. They make conjectures about the form and meaning of the solution and plan a solution pathway rather than simply jumping into a solution attempt. They consider analogous problems, and try special cases and simpler forms of the original problem in order to gain insight into its solution. They monitor and evaluate their progress and change course if necessary. Older students might, depending on the context of the problem, transform algebraic expressions or change the viewing window on their graphing calculator to get the information they need. Mathematically proficient students can explain correspondences between equations, verbal descriptions, tables and graphs or draw diagrams of important features and relationships, graph data and search for regularity or trends. Younger students might rely on using concrete objects or pictures to help conceptualize and solve a problem. Mathematically proficient students check their answers to problems using a different method, and they continually ask themselves, "Does this make sense?" They can understand the approaches of others to solving complex problems and identify correspondences between different approaches.

Math Practice 2: Reason abstractly and quantitatively.

Mathematically proficient students make sense of quantities and their relationships in problem situations. They bring two complementary abilities to bear on problems involving quantitative relationships: the ability to decontextualize (to abstract a given situation and represent it symbolically and manipulate the representing symbols as if they have a life of their own, without necessarily attending to their referents) and the ability to contextualize (to pause as needed during the manipulation process in order to probe into the referents for the symbols involved). Quantitative reasoning entails habits of creating a coherent representation of the problem at hand; considering the units involved; attending to the meaning of quantities, not just how to compute them; and knowing and flexibly using different properties of operations and objects.

Math Practice 3: Construct viable arguments and appreciate and critique the reasoning of others.

Mathematically proficient students understand and use stated assumptions, definitions and previously established results in constructing arguments. They make conjectures and build a logical progression of statements to explore the truth of their conjectures. They are able to analyze situations by breaking them into cases and can recognize and use counterexamples. They justify their conclusions, communicate them to others and respond to the arguments of others. They reason inductively about data, making plausible arguments that take into account the context from which the data arose. Mathematically proficient students are also able to compare the effectiveness of two plausible arguments, distinguish correct logic or reasoning from that which is flawed and, if there is a flaw in an argument, explain what it is. Elementary students can construct arguments using concrete referents such as objects, drawings, diagrams and actions. Such arguments can make sense and be correct, even though they are not generalized or made formal until later grades. Later, students learn to determine domains to which an argument applies. While communicating their own mathematical ideas is important, students also learn to be open to others' mathematical ideas. They appreciate a different perspective or approach to a problem and learn how to respond to those ideas, respecting the reasoning of others (Gutiérrez 2017, 17–18)[3]. Together, students make sense of the mathematics by asking helpful questions that clarify or deepen everyone's understanding.

Math Practice 4: Model with mathematics.

Mathematically proficient students can apply the mathematics they know to solve problems arising in everyday life, society and the workplace. In early grades, this might be as simple as writing an addition equation to describe a situation. In middle grades, a student might apply proportional reasoning to plan a school event or analyze a problem in the community. By high school, a student might use geometry to solve a design problem or use a function to describe how one quantity of interest depends on another. Mathematically proficient students who can apply what they know are comfortable making assumptions and approximations to simplify a complicated situation, realizing that these may need revision later. They are able to identify important quantities in a practical situation and map their relationships using such tools as diagrams, two-way tables, graphs, flowcharts and formulas. They can analyze those relationships

[3] Gutiérrez, Rochelle. 2017. "Living Mathematx: Towards a Vision for the Future." Philosophy of Mathematics Education Journal. November: 2-26. https://eric.ed.gov/?id=ED581384

mathematically to draw conclusions. They routinely interpret their mathematical results in the context of the situation and reflect on whether the results make sense, possibly improving the model if it has not served its purpose.

Math Practice 5: Use appropriate tools strategically.

Mathematically proficient students consider the available tools when solving a mathematical problem. These tools might include pencil and paper, concrete models, a ruler, a protractor, a calculator, a spreadsheet, a computer algebra system, a statistical package or dynamic geometry software. Proficient students are sufficiently familiar with tools appropriate for their grade or course to make sound decisions about when each of these tools might be helpful, recognizing both the insight to be gained and their limitations. For example, mathematically proficient high school students analyze graphs of functions and solutions generated using a graphing calculator. They detect possible errors by strategically using estimation and other mathematical knowledge. When making mathematical models, they know that technology can enable them to visualize the results of varying assumptions, explore consequences and compare predictions with data. Mathematically proficient students at various grade levels are able to identify relevant external mathematical resources, such as digital content located on a website, and use them to pose or solve problems. They are able to use technological tools to explore and deepen their understanding of concepts.

Math Practice 6: Attend to precision.

Mathematically proficient students try to communicate precisely to others. They try to use clear definitions in discussion with others and in their own reasoning. They state the meaning of the symbols they choose, including using the equal sign consistently and appropriately. They are careful about specifying units of measure and labeling axes to clarify the correspondence with quantities in a problem. They calculate accurately and efficiently and express numerical answers with a degree of precision appropriate for the problem context. In the elementary grades, students give carefully formulated explanations to each other. By the time they reach high school they have learned to examine claims and make explicit use of definitions.

Math Practice 7: Look for and make use of structure.

Mathematically proficient students look closely to discern a pattern or structure. Young students, for example, might notice that three and seven more is the same amount as seven and three more, or they may sort a collection of shapes according to how many sides the shapes have. Later, students will see 7 × 8 equals the well-remembered 7 × 5 + 7 × 3, in preparation for learning about the distributive property. In the expression $x^2 + 9x + 14$, older students can see the 14 as 2 × 7 and the 9 as 2 + 7. They recognize the significance of an existing line in

a geometric figure and can use the strategy of drawing an auxiliary line for solving problems. They also can step back for an overview and shift perspective. They can see complicated things, such as some algebraic expressions, as single objects or as being composed of several objects. For example, they can see $5 - 3(x - y)^2$ as 5 minus a positive number times a square and use that to realize that its value cannot be more than 5 for any real numbers x and y.

Math Practice 8: Look for and express regularity in repeated reasoning.

Mathematically proficient students notice if calculations are repeated and look both for general methods and for shortcuts. Upper elementary students might notice when dividing 25 by 11 that they are repeating the same calculations over and over again, and conclude they have a repeating decimal. By paying attention to the calculation of slope as they repeatedly check whether points are on the line through (1, 2) with slope 3, middle school students might abstract the equation $\frac{(y-2)}{(x-1)} = 3$. Noticing the regularity in the way terms cancel when expanding $(x - 1)(x + 1)$, $(x - 1)(x^2 + x + 1)$ and $(x - 1)(x^3 + x^2 + x + 1)$ might lead them to the general formula for the sum of a geometric series. As they work to solve a problem, mathematically proficient students maintain oversight of the process, while attending to the details. They continually evaluate the reasonableness of their intermediate results.